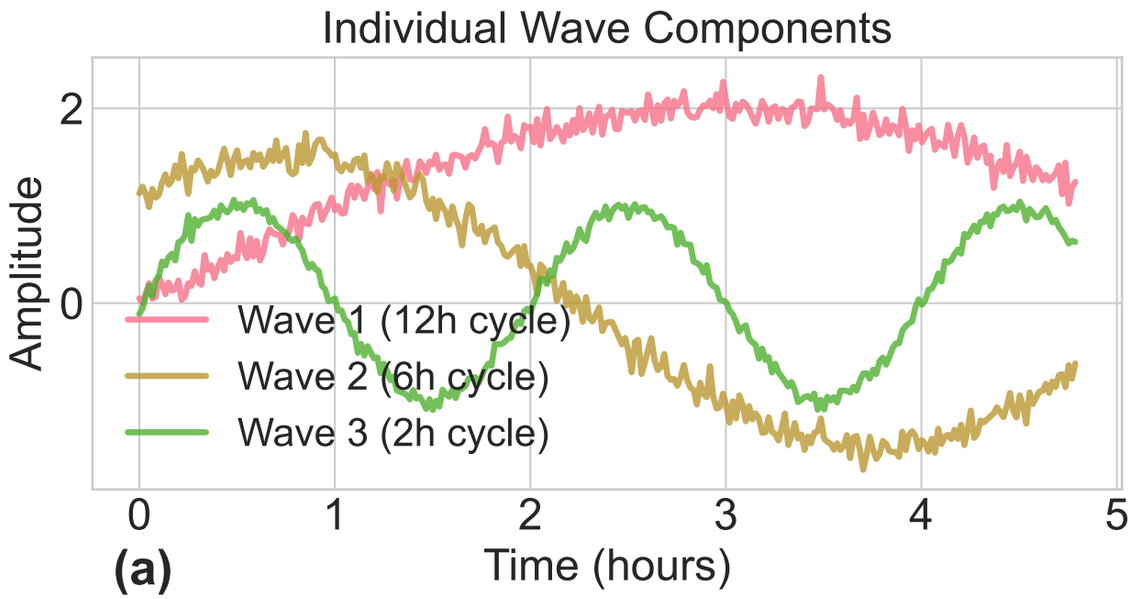
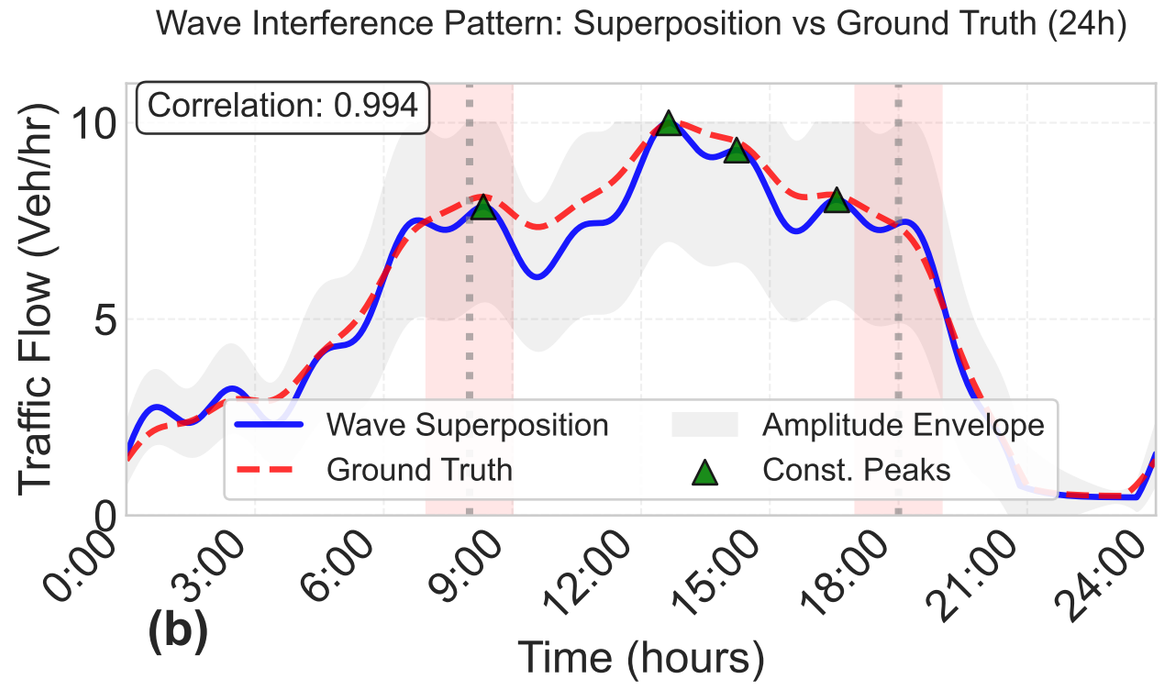
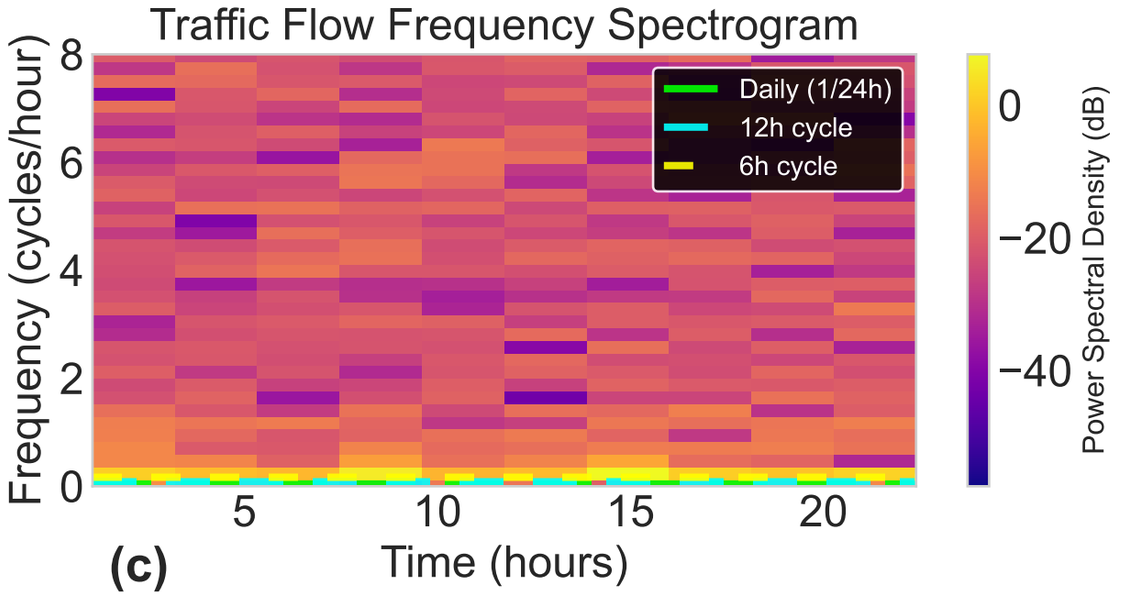
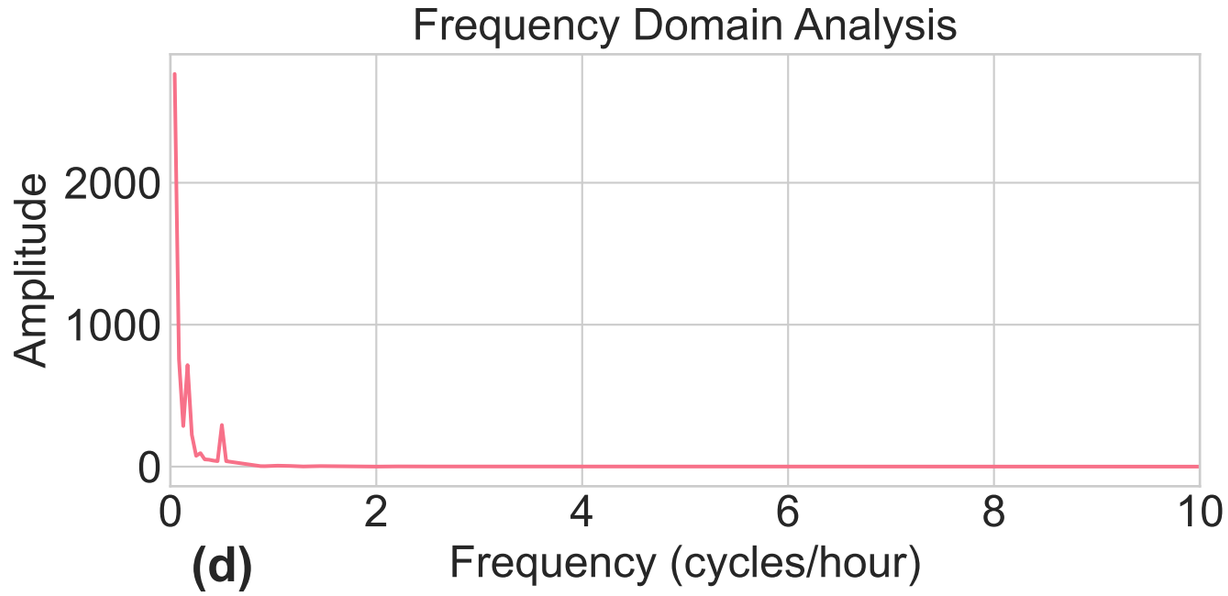
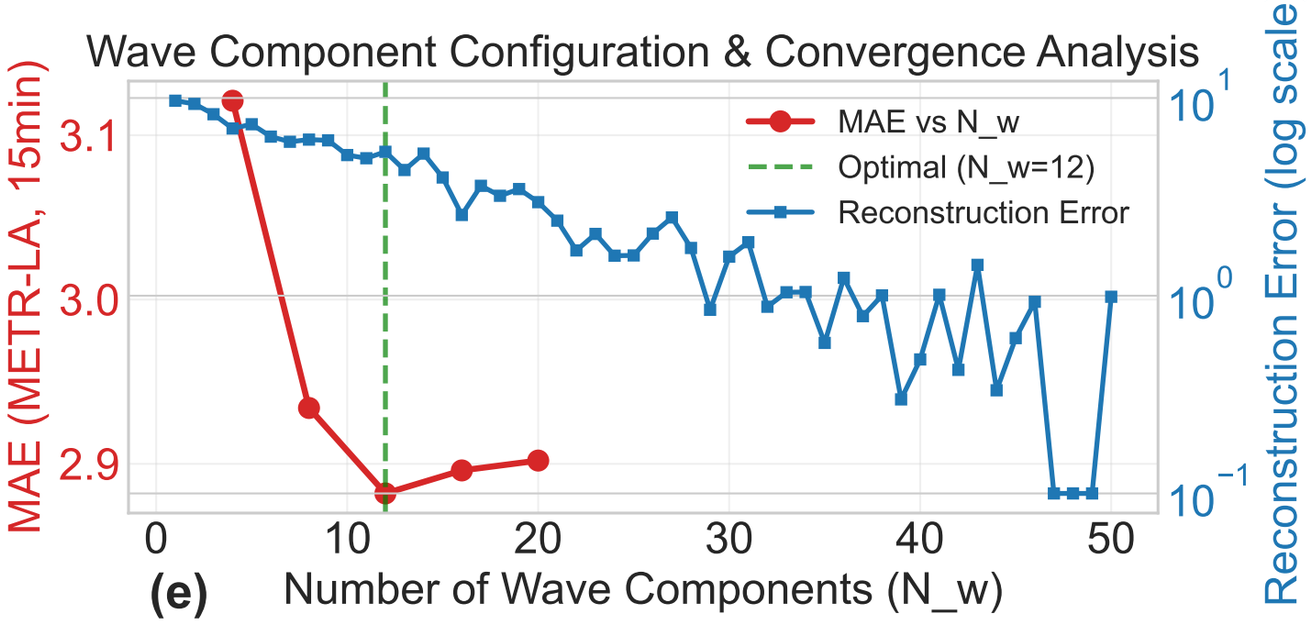
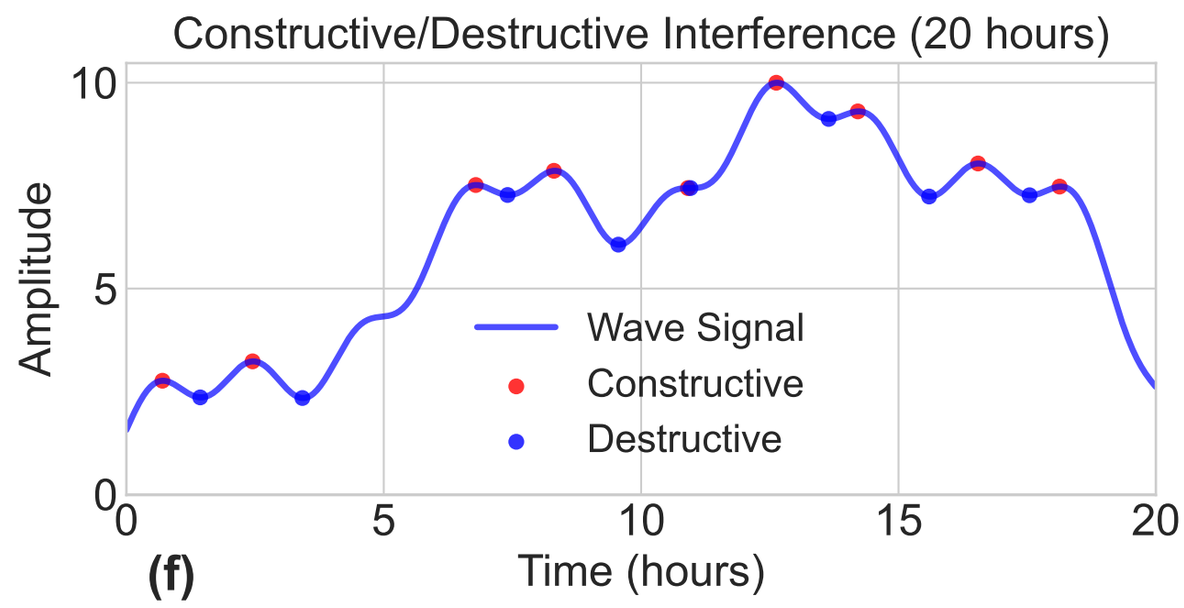

Individual Wave Components
Amplitude
Wave 1 (12h cycle)
Wave 2 (6h cycle)
Wave 3 (2h cycle)
Time (hours)
(a)


Wave Interference Pattern: Superposition vs Ground Truth (24h)
Correlation: 0.994
Traffic Flow (Veh/hr)
Wave Superposition
Ground Truth
Amplitude Envelope
Const. Peaks
Time (hours)
(b)


Traffic Flow Frequency Spectrogram
Frequency (cycles/hour)
Daily (1/24h)
12h cycle
6h cycle
Power Spectral Density (dB)
Time (hours)
(c)


Frequency Domain Analysis
Amplitude
Frequency (cycles/hour)
(d)


Wave Component Configuration & Convergence Analysis
MAE (METR-LA, 15min)
MAE vs N_w
Optimal (N_w=12)
Reconstruction Error
Reconstruction Error (log scale)
Number of Wave Components (N_w)
(e)


Constructive/Destructive Interference (20 hours)
Amplitude
Wave Signal
Constructive
Destructive
Time (hours)
(f)